Dr. Nina Miolane

I study the mathematical principles that enable intelligence in brains and machines.

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Summary

My research sits at the intersection of mathematics, artificial intelligence, and neuroscience. I use mathematics to unify the study of intelligence in brains and machines. In neuroscience, we still do not know how large populations of neurons give rise to perception, memory, and learning. In AI/ML, we face significant challenges to understand systems of our own making, leaving them hard to control and trust. My hypothesis is that we lack appropriate mathematics that reveals the principles of intelligence in brains and machines. My research develops the missing framework, uses it to formalize a theory of intelligence across substrates, and builds it into novel artificial neural networks to deliver superior performance without necessarily scaling data, parameters, or compute.

Education

2018-2020 Postdoctoral Fellow and Lecturer in Statistics at **Stanford** (U.S.)
Advised by Susan Holmes
- *Inria Silicon Valley Postdoctoral Fellowship (ranked 1st)*
- *Stanford's CONNECT award*
- *1st Prize ($100,000) in the C3.ai Covid-19 Grand Challenge*

2013-2016 Ph.D. in Geometric Statistics at **Inria** with co-supervision at **Stanford** (France, U.S.)
Advised by Xavier Pennec, Co-Supervised by Susan Holmes
- *France-Stanford Center for Interdisciplinary Studies Fellowship (ranked 1st)*
- *Inria S-Cordi National Fellowship (ranked 1st)*
- *1st Prize at MICCAI Educational Challenge, MIT, Boston*
- *L'Oréal-Unesco For Women in Science National Prize (ranked 1st computer scientist)*

2012-2013 MSc. in Mathematical Physics at **Imperial College London** (U.K.)
2010-2013 BSc. and MSc. in Mathematics and Physics at **Ecole Polytechnique** (France)
2008-2009 Classes Préparatoires in Mathematics at **Ecole Sainte Genevieve (Ginette)** (France)

Work Experience

Assistant Professor, UC Santa Barbara (UCSB) 2021 - present
- Principal Investigator, Geometric Intelligence Lab: Lead team of 16 researchers
- Co-Director of the REAL AI for Science Initiative at UCSB
- Co-Director of the AI Core of the Bowers Women's Brain Health Initiative
- *A $45M endowed center with $10M dedicated to the AI Core under my co-direction*

Research Software Engineer, Atmo, San Francisco 2020 - 2021
- Built large-scale machine learning models and infrastructure in AI for science.

Research Software Engineer, Caption Health, San Francisco 2017 - 2018
- Built AI-guided ultrasound probe for cardiac imaging.
- *Delivered Nation's 1st FDA-cleared AI-guided medical device*

Selected Grants and Gifts

I received 11 grants and gifts totaling \$3.8M (PI share), including a NSF CAREER award, a NIH R01, and an NSF MFAI as sole and lead PI (x2), respectively.

I co-founded the \$45M endowed Bowers Women's Brain Health Initiative, headquartered at UCSB and spanning 9 universities nationwide (7 UCs, Stanford, Cornell), with \$10M dedicated to the AI Core under my co-direction.

2026 NSF MFAI. *Discovering Higher-Order Scientific Laws With Deep Learning* **Lead PI**
Total \$1M · Share \$300k

2025 Anonymous Gift. *REAL AI for Science* **Lead PI**
Total \$500k · Share \$166k

2025 Arlequin AI Gift. *Topological Deep Learning* **Sole PI**
Total \$100k · Share \$100k

2025 Chan-Zuckerberg Initiative Gift. *Advancing Inclusive AI in Biomedical Datasets* **Co-PI**
Total \$1.47M · Share \$70k

2024 Noyce Foundation Gift. *AI for Women's Brain Health* **Sole PI**
Total \$600k · Share \$600k

2024 Chan-Zuckerberg Initiative Gift. *AI Model of the Maternal Brain* **Lead PI**
Total \$1M · Share \$500k

2023 NSF CAREER. *Advancing Shape Learning for Biosciences* **Sole PI**
Total \$496k · Share \$496k

2023 NSF. *Lie Group Representation Learning for Vision* **Co-PI**
Total \$1.2M · Share \$500k

2022 Noyce Initiative. *Leveraging AI to Advance Women's Health* **Co-PI**
Total \$918k · Share \$320k

2021 NSF. *Deep Learning Framework Using Cell Complex Neural Networks* **Co-PI**
Total \$547k · Share \$335k

2021 NIH R01. *Learning Conformations of Membrane Proteins from Cryo-EM* **Lead PI**
Total \$600k · Share \$464k

Total \$8.43M · Share \$3.8M

Selected Awards

1. Named to TIME's Best Inventions for *HerBrain*: digital twin of the brain across pregnancy (2025).
2. Public Voices Fellow of the OpEd Project (2025).
3. Hellman Fellow (2023).
4. NSF CAREER Award (2023).
5. 1st Prize (\$100,000) in the C3.ai COVID-19 Grand Challenge (2020).
6. L'Oréal–UNESCO For Women in Science National Prize, 1st computer scientist (2016).

Publications

My research explores how the mathematics of geometry, topology, and symmetry governs learning and computation in brains and machines.

This work comprises **39 peer-reviewed conference and journal papers**. A further 5 appeared in peer-reviewed workshop proceedings.

Bold: candidate ***Italic Bold***: advised Ph.D. student, visiting Ph.D. student, postdoc, or alumni *: equal contribution

AI/ML Conferences

[1] ***Cornelis***, ***Bernardez***, Jeong, **Miolane**. "When Machine Learning Gets Personal: Evaluating Prediction and Explanation". In: *International Conference on Learning Representations (ICLR)*. 2026.

[2] ***Langendonck***, ***Bernardez***, **Miolane**, Barlet-Ros. "GraphUniverse: Enabling Systematic Evaluation of Inductive Generalization". In: *International Conference on Learning Representations (ICLR)*. 2026.

[3] Marchetti*, ***Kunin****, ***Myers***, ***Acosta***, **Miolane**. "Sequential Group Composition: A Window into the Mechanics of Deep Learning". In: *International Conference on Machine Learning (ICML)*. 2026.

[4] ***Papillon****, ***Bernardez****, Barreiro, ***Montagna***, Devaux, Jardin, **Miolane**. "Look Before You Lift: Visual and Quantitative Diagnostics for Topological Deep Learning". In: *Proceedings of Machine Learning Research (PMLR)*. 2026.

[5] ***Kunin***, Marchetti, Chen, Karkada, Simon, DeWeese, Ganguli, **Miolane**. "Alternating Gradient Flows: A Theory of Feature Learning in Two-layer Neural Networks". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2025.

[6] ***Kurtkaya****, ***Dinc****, Yuksekgonul, Blanco-Pozo, Cirakman, Schnitzer, Yemez, Tanaka, Yuan, **Miolane**. "Dynamical phases of short-term memory mechanisms in RNNs". In: *International Conference on Machine Learning (ICML)*. 2025.

[7] ***Papillon***, ***Bernardez***, Battiloro, **Miolane**. "TopoTune: A Framework for Generalized Combinatorial Complex Neural Networks". In: *International Conference on Machine Learning (ICML)*. 2025.

[8] ***Acosta***, ***Dinc***, Redman, Madhav, ***Klindt***, **Miolane**. "Global Distortions from Local Rewards: Neural Coding Strategies in Path-Integrating Neural Systems". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2024.

[9] ***Mataigne***, Mathe, ***Sanborn***, Hillar, **Miolane**. "The Selective G-Bispectrum and its Inversion: Applications to G-Invariant Networks". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2024.

[10] Papamarkou, Birdal, Bronstein, Carlsson, Curry, Gao, Hajij, Kwitt, Lio, Di Lorenzo, Maroulas, **Miolane**, Nasrin, Natesan Ramamurthy, Rieck, Scardapane, Schaub, Veličković, Wang, Wang, Wei, Zamzmi. "Position: Topological Deep Learning is the New Frontier for Relational Learning". In: *International Conference on Machine Learning (ICML)*. 2024.

[11] Redman, ***Acosta***, Acosta–Mendoza, **Miolane**. "Not so griddy: Internal representations of RNNs path integrating more than one agent". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2024.

[12] ***Sanborn***, **Miolane**. "A general framework for robust g-invariance in g-equivariant networks". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023.

[13] Levy, Poitevin, Martel, Nashed, Peck, **Miolane**, Ratner, Dunne, Wetzstein. "CryoAI: Amortized inference of poses for ab initio reconstruction of 3D molecular volumes from real cryo-EM images". In: *European Conference on Computer Vision (ECCV)*. 2022.

[14] **Miolane**, Holmes. "Learning Weighted Submanifolds with Variational Autoencoders and Riemannian Variational Autoencoders". In: *Conference on Computer Vision and Pattern Recognition (CVPR)*. 2020.

Applied Math Conferences

[1] ***Kushner***, Modi, **Miolane**. "Learning Riemannian Metrics for Interpolating Animations". In: *Conference on Geometric Science of Information (GSI)*. **Oral**. 2025.

[2] ***Mataigne***, Absil, **Miolane**. "On the Approximation of the Riemannian Barycenter". In: *Conference on Geometric Science of Information (GSI)*. **Oral**. 2025.

[3] Li, Prasad, **Miolane**, Dao Duc. "Using a Riemannian Elastic Metric for Statistical Analysis of Tumor Cell Shape Heterogeneity". In: *Conference on Geometric Science of Information (GSI)*. **Oral**. 2023.

[4] ***Shewmake***, **Miolane**, Olshausen. "Group Equivariant Sparse Coding". In: *Conference on Geometric Science of Information (GSI)*. **Oral**. 2023.

AI/ML and Applied Math Journals

[1] ***Klindt***, O'Neill, Reizinger, Maurer, **Miolane**. "From superposition to sparse codes: interpretable representations in neural networks". In: *Nature Machine Intelligence* (2026).

[2] Baracaldo, King, Yan, Lin, **Miolane**, Gu. "Unsupervised Cell Segmentation by Fast Gaussian Processes". In: *The New England Journal of Statistics in Data Science* (2026).

[3] ***Mataigne***, Absil, **Miolane**. "Bounds on the geodesic distances on the Stiefel manifold for a family of Riemannian metrics". In: *Linear Algebra and Its Applications* (2026).

[4] ***Papillon****, ***Sanborn****, Mathe*, Cornelis*, ***Bertics***, Buracas, Lillemark, ***Shewmake***, ***Dinc***, Pennec, **Miolane**. "Beyond Euclid: an illustrated guide to modern machine learning with geometric, topological, and algebraic structures". In: *Machine Learning: Science and Technology (MLST)* (2025).

[5] ***Pereira***, Le Brigant, ***Myers***, Hartman, Khan, Tuerkoen, Dold, Gu, Suárez-Serrato, **Miolane**. "Learning from landmarks, curves, surfaces, and shapes in Geomstats". In: *ACM Transactions on Mathematical Software (TOMS)* (2025).

[6] Telyatnikov*, ***Bernardez****, ***Montagna***, Vasylenko, Zamzmi, Hajij, Schaub, **Miolane**, Scardapane, Papamarkou. "TopoBench: A framework for benchmarking topological deep learning". In: *Journal of Data-centric Machine Learning Research (DMLR)* (2025).

[7] Hajij*, ***Papillon****, Frantzen*, Agerberg, AlJabea, Ballester, Battiloro, ***Bernardez***, Birdal, Brent, Chin, Escalera, Fiorellino, Gardaa, Gopalakrishnan, Govil, Hoppe, Karri, Khouja, Lecha, Livesay, Meißner, Mukherjee, Nikitin, Papamarkou, Prílepok, Ramamurthy, Rosen, Guzmán-Sáenz, Salatiello, Samaga, Scardapane, Schaub, Scofano, Spinelli, Telyatnikov, Truong, Walters, Yang, Zaghen, Zamzmi, Zia, **Miolane**. "TopoX: a suite of Python packages for machine learning on topological domains". In: *Journal of Machine Learning Research (JMLR)* (2024).

[8] ***Mataigne***, Zimmermann, **Miolane**. "An efficient algorithm for the Riemannian logarithm on the Stiefel manifold for a family of Riemannian metrics". In: *SIAM Journal on Matrix Analysis and Applications* (2024).

[9] Le Brigant, Deschamps, Collas, **Miolane**. "Parametric information geometry with Geomstats". In: *ACM Transactions on Mathematical Software (TOMS)* (2023).

[10] Guigui, **Miolane**, Pennec. "Introduction to Riemannian Geometry and Geometric Statistics: from basic theory to implementation with Geomstats". In: *Foundations and Trends in Machine Learning (FTML)* (2023).

[11] ***Utpala***, Vepakomma, **Miolane**. "Differentially Private Fréchet Mean on the Manifold of Symmetric Positive Definite Matrices with log-Euclidean Metric". In: *Transactions on Machine Learning Research (TMLR)* (2023).

[12] **Miolane**, Guigui, Le Brigant, Mathe, Hou, Thanwerdas, Heyder, Peltre, Koep, Zaatiti, Hajri, Cabanes, Gerald, Chauchat, ***Shewmake***, Brooks, Kainz, Donnat, Holmes, Pennec. "Geomstats: A Package for Riemannian Geometry in Machine Learning". In: *Journal of Machine Learning Research (JMLR)* (2020).

[13] **Miolane**, Holmes, Pennec. "Topologically Constrained Template Estimation via Morse–Smale Complexes Controls Its Statistical Consistency". In: *SIAM Journal on Applied Algebra and Geometry* (2018).

[14] **Miolane**, Holmes, Pennec. "Template shape estimation: correcting an asymptotic bias". In: *SIAM Journal on Imaging Sciences* (2017).

[15] **Miolane**, Pennec. "Computing Bi-Invariant Pseudo-Metrics on Lie Groups for Consistent Statistics". In: *Entropy* (2015).

Applied Science Journals

[1] ***Bertics***, Chrastil, **Miolane**, Carlson. "Flow Tree: A Model for Navigator Dynamics and Environmental Complexity". In: *Journal of the Royal Society Interface* (2026). In Press.

[2] Wood, Rabin, Caunca, Iadipaolo, ***Cornelis***, **Miolane**, Pham, Borger, Diaz, Paolillo, Kramer, Pritschet, Taylor, Panizzon, Reyes, Denkinger, Ashton, Johnson, Jacobs, Saloner, Casaletto. "Blood-based proteomic signatures of spontaneous menopause: Implications for later-life brain aging and Alzheimer's disease risk". In: *Nature Medicine* (2026). Accepted for Publication.

[3] ***Klindt***, Hyvärinen, Levy, **Miolane**, Poitevin. "Towards interpretable Cryo-EM: disentangling latent spaces of molecular conformations". In: *Frontiers in Molecular Biosciences* (2024).

[4] Li, Prasad, **Miolane**, Dao Duc. "Unveiling cellular morphology: statistical analysis using a Riemannian elastic metric in cancer cell image datasets". In: *Information Geometry* (2024).
[5] Gabbert, Campanale, Mondo, Mitchell, ***Myers***, Streichan, **Miolane**, Montell. "Septins regulate border cell surface geometry, shape, and motility downstream of Rho in Drosophila". In: *Developmental Cell* (2023).
[6] Li, Mirone, Prasad, **Miolane**, Legrand, Dao Duc. "Orthogonal outlier detection and dimension estimation for improved MDS embedding of biological datasets". In: *Frontiers in Bioinformatics* (2023).
[7] Donnat, Levy, Poitevin, Zhong, **Miolane**. "Deep generative modeling for volume reconstruction in cryo-electron microscopy". In: *Journal of Structural Biology* (2022).

AI/ML Workshops (with Proceedings)

[1] ***Kurtkaya***, Harmanli*, Cimen*, Alexander, **Miolane**, ***Dinc***, Yemez. "Learning rate collapse prevents training recurrent neural networks at scale". In: *NeurIPS NeurReps Workshop on Symmetry and Geometry in Neural Representations (NeurIPSW)*. 2025.
[2] ***Myers***, **Miolane**. "On Accuracy and Speed of Geodesic Regression: Do Geometric Priors Improve Learning on Small Datasets?" In: *CVPR Workshop on Learning with Limited Labeled Data for Image and Video Understanding (CVPRW)*. 2024.
[3] ***Acosta***, ***Sanborn***, Dao Duc, Madhav, **Miolane**. "Quantifying Extrinsic Curvature in Neural Manifolds". In: *CVPR Workshop on Topology, Algebra and Geometry (CVPRW)*. 2023.
[4] ***Myers***, Taylor, Jacobs, **Miolane**. "Geodesic Regression Characterizes 3D Shape Changes in the Female Brain During Menstruation". In: *ICCV Workshop on Computer Vision for Automated Medical Diagnosis (ICCVW)*. 2023.
[5] ***Shewmake***, Buracas, Lillemark, Shin, Bekkers, **Miolane**, Olshausen. "Visual Scene Representation with Hierarchical Equivariant Sparse Coding". In: *NeurIPS NeurReps Workshop on Symmetry and Geometry in Neural Representations (NeurIPSW)*. 2023.

Patents

- Hajij, Alzamzmi, and **Miolane**. "Multimodal cell complex neural networks for prediction of multiple drug side effects severity and frequency," U.S. Patent 12051512 (2024).
- Cadieu, Hong, Koepsell, Mathe, Poilvert, Cannon, Romano, Bilenko, Chen, and **Miolane**. "Video clip selector for medical imaging and diagnosis," U.S. Patent 10631791 (2020). Continuations 11166678, 11497451.

Open-source Software

I create and maintain open-source scientific infrastructure adopted by the geometric and topological deep learning communities, with three flagship libraries (2,000+ combined GitHub stars) and peer-reviewed software papers in *JMLR* (2020, 2024) and *DMLR* (2025).

Geomstats GitHub · **1.5k stars** · **286 forks**

- Creator and developer. Paper in *JMLR* (2020). Follow-ups in *FTML* (2023), *TOMS* (2023, 2025).
- Organizer, ICLR Comp. Geometry & Topology Challenges (2021, 2022) — 2022 results in *PMLR*.
  - 27 and 33 international contributors (organizers, reviewers and participants).

TopoX GitHub · **334 stars** · **103 forks**

- Co-creator and developer (TopoNetX, TopoEmbedX, TopoModelX). Paper in *JMLR* (2024).
- Senior co-organizer, ICML Topological Deep Learning Challenges (2023, 2024) — results in *PMLR*.
  - 52 and 73 international contributors (organizers, reviewers and participants).

TopoBench GitHub · **235 stars** · **77 forks**

- Developer and senior co-mentor. Paper in *DMLR* (2025).
- Senior co-mentor, TAG Topological Deep Learning Challenges (2025, 2026) — 2025 results in *PMLR*.
  - 66 international contributors (organizers, reviewers and participants).

Talks

I gave 51 invited talks since 2022 at international universities, research institutes, and scholarly conferences and workshops, including 1 distinguished lecture, 2 keynotes, and 5 workshop keynotes. Additionally, I've given 18 invited talks within UCSB.

Distinguished Lecture, Keynotes & Workshop Keynotes

1. **Opening Keynote**, Geometric Science of Information (GSI), Saint-Malo. *Topological Deep Learning* (2025).
2. **Keynote**, NeurIPS Workshop on New Perspectives in Graph ML. *Topological Deep Learning* (2025).
3. **Keynote**, NeurIPS Workshop Mouse vs AI. *The Algebra of Spatial Navigation* (2025).
4. **Annual Tutte Lecture, Distinguished Speaker**, Ottawa. *Geometric Computations in Natural and Artificial Brains* (2024).
5. **Closing Keynote**, CZI Imaging The Future Annual Event. *From Nano to Macro: Unlocking the Brain with AI-Driven Imaging Insights* (2024).
6. **Keynote**, CVPR Workshop on Equivariance in Vision. *Hierarchical Equivariance in Artificial and Natural Brains* (2024).
7. **Keynote**, CVPR Workshop on Topology, Algebra and Geometry. *Architectures of Topological Deep Learning* (2023).
8. **Keynote**, CVPR Workshop on Deep Learning for Geometric Computing. *Shape Learning in Biomedical Imaging* (2022).

Invited Talks at External Venues

1. **UC Berkeley, Redwood Center for Theoretical Neuroscience**. *Geometric Intelligence in Brains and Machines* (August 2026).
2. **Goodfire**. *Geometric Intelligence in Brains and Machines* (August 2026).
3. **Long Now Foundation, San Francisco**. *The Geometry of Consciousness* (April 2026).
4. **International Winter Neuroscience Conference**. *Symmetry in Brains and Machines* (April 2026).
5. **University of Texas, Dallas' School of Behavioral and Brain Sciences, Center for Brain Health**. *Digital Twin Brain to Catapult Precision Brain Health* (February 2026).
6. **McMaster University Seminar**. *Topological Deep Learning* (October 2025).
7. **Montecito Ideas Club**. *Healthy and Pathological Aging: Unlocking the Brain with AI. (Outreach)* (October 2025).
8. **Cold Spring Harbor Laboratory Seminar Series**. *Geometric Intelligence in Minds and Machines* (July 2025).
9. **Stanford University, Workshop Bridging Biology and Statistics In Honor of Professor Holmes**. *The Role of AI in Advancing Women's Brain Health* (June 2025).
10. **NSF-Simons National Institute for Mathematics and Theory in Biology (NITMB), Workshop New Developments in the Theory and Methodology of Graph Neural Networks**. *Topological Deep Learning* (April 2025).
11. **Instituto de Matemáticas, Geometric Intelligence Workshop**. *Geometric Intelligence in Minds and Machines* (April 2025).
12. **COSYNE Workshop on GNNs and Graphical Analysis in Neuroscience**. *A Survey of Message Passing Topological Neural Networks* (March 2025).
13. **University of Texas, Dallas' School of Behavioral and Brain Sciences, Center for Brain Health**. *AI and the Digital Twin Brain* (February 2025).
14. **NeurReps Seminar Series**. *Geometric Intelligence in Minds and Machines* (February 2025).
15. **Chan-Zuckerberg Initiative Seminar**. *Leveraging AI for Women's Brain Health Research* (February 2025).
16. **Erwin Schrödinger International Institute**. *G-Invariance in G-Equivariant Networks* (February 2025).

17. **Noyce Foundation Annual Event**. *The Role of AI in Advancing Women's Brain Health* (October 2024).
18. **Cognitive Computational Neuroscience Conference, Battle of the Metrics Community Event**. *Riemannian Geometry of Neural Representations in Natural and Artificial Intelligence* (August 2024).
19. **ICML Workshop on Geometry-grounded Representation Learning and Generative Modeling**. *Geometric Intelligence* (July 2024).
20. **Eresfjord School in Mathematical Methods in Computational Neuroscience**. *Hierarchical Equivariance in Artificial and Natural Brains* (July 2024).
21. **MIT DHIVE Program**. *Digital Twins and the Role of AI in Advancing Women's Brain Health* (June 2024).
22. **UC Riverside Seminar**. *Robust G-Invariance in G-Equivariant Networks* (May 2024).
23. **Bowers Women's Brain Health Seminar**. *Digital Twins and the Role of AI in Advancing Women's Brain Health* (May 2024).
24. **Caltech's Math & Machine Learning Seminar**. *Robust G-Invariance in G-Equivariant Networks* (February 2024).
25. **University of Chicago, Institute for Mathematical and Statistical Innovation, Workshop on Graph Neural Networks for the Sciences**. *A Survey of Message Passing Topological Neural Networks* (January 2024).
26. **Harvard University, Department of Applied Mathematics**. *The Differential Geometry of Neural Manifolds* (November 2023).
27. **University of North Carolina at Chapel Hill, Applied Physical Sciences Colloquium**. *Architectures of Topological Deep Learning: A Survey of Topological Neural Networks* (November 2023).
28. **MIT Summer School of Geometry**. *Architectures of Topological Deep Learning: A Survey of Topological Neural Networks* (August 2023).
29. **Eresfjord School in Mathematical Methods in Computational Neuroscience**. *Geometry and Topology of Neural Manifolds* (July 2023).
30. **University of British Columbia, Center for Brain Health**. *Architectures of Topological Deep Learning: A Survey of Topological Neural Networks* (June 2023).
31. **CVPR Workshop on Computational Cameras and Displays**. *Deep Generative Modeling for Volume Reconstruction in Cryo-Electron Microscopy: A Survey* (June 2023).
32. **Flatiron Institute, Simons Foundation, CryoEM Summer Workshop**. *Deep Generative Modeling for Volume Reconstruction in Cryo-Electron Microscopy* (June 2023).
33. **UCLA Applied Mathematics Seminar**. *Coding Differential Geometry for Machine Learning* (May 2023).
34. **IEEE Central Coast Event Talk**. *Geometric Learning for Shape Analysis from Bioimaging Data* (April 2023).
35. **Data Science Initiative at SISSA**. *Bioshape Analysis with Geometric Learning* (January 2023).
36. **INRIA Saclay, Seminar of the SODA and MIND (previously Parietal) teams**. *Geomstats: A Python Package for Geometry in Machine Learning* (October 2022).
37. **Institut Poincaré, Workshop on Geometry, Topology in Statistics and Data Science**. *Geomstats: A Python Package for Geometry in Machine Learning* (October 2022).
38. **SIAM Conference on Mathematics of Data Science, Mini Symposium of Statistics and Machine Learning in Topological and Geometric Data Analysis**. *Geomstats: A Python Package for Differential Geometry in Machine Learning* (September 2022).
39. **Geo2Int (Geometry and Geometric Integration) Workshop**. *Geomstats: A Python Package for Differential Geometry in Machine Learning* (September 2022).
40. **University of Washington, eScience Institute Seminar**. *Geometric Learning for Shape Analysis in Bioimaging* (June 2022).
41. **Sage Math Conference**. *Geomstats: A Python Package for Differential Geometry in Machine Learning* (June 2022).
42. **Pacific Institute for the Mathematical Sciences (PIMS) Workshop**. *CompSPI: Open-Source Software for Biological Imaging* (May 2022).
43. **Johns Hopkins University, Center for Imaging Science Workshop**. *Geometric Learning for Shape Analysis from Bioimaging* (May 2022).

Invited Talks at UCSB

1. **UCSB AI Library**. *AI-Powered Science and the Digital Twin Revolution. (Outreach)* (May 2026).
2. **UCSB Physics Colloquium**. *Symmetry in Brains and Machines* (March 2026).
3. **UCSB REAL AI for Science Initiative**. *AI-Powered Science and the Digital Twin Revolution. (Outreach)* (February 2026).
4. **UCSB Spatial Navigation**. *The Geometry and Algebra of Spatial Navigation* (January 2026).
5. **UCSB Center for Aging and Longevity**. *Direct Relief, Santa Barbara. Healthy and Pathological Aging: Unlocking the Brain with AI. (Outreach)* (June 2025).
6. **UCSB Physics Graduate Seminar**. *From Theoretical Physics to Geometric Intelligence* (February 2024).
7. **UCSB Workshop Spatial Data Science in an Age of Scientific Disruption**. *Geometric Artificial Intelligence. (Outreach)* (December 2023).
8. **UCSB ECE Summit**. *Advancing Biomedicine with AI for Shape Analysis* (April 2023).
9. **UCSB WISE Panels**. *Women in Science & Engineering: Academic Careers (Outreach)* (April 2023).
10. **UCSB Differential Geometry Seminar**. *Geomstats: Coding Differential Geometry for Machine Learning* (February 2023).
11. **UCSB Physics Graduate Seminar**. *Riemannian Geometry: From General Relativity to Biomedical Imaging* (February 2023).
12. **UCSB Machine Learning Seminar**. *Geometric Machine Learning* (January 2023).
13. **UCSB ECE Graduate Seminar**. *Geometric Machine Learning* (November 2022).
14. **UCSB Cryo-EM FIB Forum**. *Deep Generative Models for Molecular Reconstructions in Cryo-Electron Microscopy* (April 2022).
15. **UCSB Psych & Brain Sciences Seminar**. *Brain Shapes* (March 2022).
16. **UCSB CS Summit**. *Biomedical Shape Analysis with Geometric Learning* (March 2022).
17. **UCSB Materials Research Outreach Symposium**. *Shape Learning: From Images to Scientific Insights. (Outreach)* (January 2022).
18. **UCSB Physics Colloquium**. *Riemannian Geometry: From General Relativity to Biomedical Imaging* (January 2022).

Mentoring

I lead a team of 16 in the Geometric Intelligence Lab: 5 sole-advised Ph.D. students, 4 co-advised at UCSB and peer institutions, 2 visiting Ph.D. students, 3 postdoctoral fellows, and 2 staff. My three postdoctoral alumni hold faculty, research-faculty, and staff scientist positions.

Nominated by my Ph.D. students for UCSB's Best Graduate Mentor award (2023).

Ph.D. Students — Sole Advisor

1. **Mathilde Papillon** (2022–present; *graduation scheduled for Oct. 2026*).
   Fellowships: NSERC Postgraduate Doctoral (2024–27); UCSB Dean's Graduate (2021); Robert E. Bell Prize in Physics, McGill (2020).
2. **Adele Myers** (2022–present; *graduation exp. Spring 2027*).
   Fellowships: NSF GRFP (2023); NSF NRT Data-Driven Biology (2022).
3. **Francisco Acosta** (2022–present; *graduation exp. Spring 2027*).
4. **Louisa Cornelis** (2024–present; *graduation exp. 2029*).
   Fellowships: Gilliam Finalist, HHMI (2025); NSF GRFP (2024).
5. **Bariscan Kurtkaya** (2025–present; *graduation exp. 2030*).

Ph.D. Students — Co-Advised at UCSB & Peer Institutions

1. **Abby Bertics**, UC Santa Barbara (2024–present; *exp. 2029*). Co-advised with Jean Carlson.
   Fellowship: UCSB Chancellor's (2024–28).
2. **Simon Mataigne**, UC Louvain (2023–present; *exp. 2027*). Co-advised with Pierre-Antoine Absil.

3. **Christian Shewmake**, UC Berkeley (2023–present; *currently pausing PhD to lead a venture-backed startup*). Co-advised with Bruno Olshausen.
4. **Liam Storan**, Stanford (2024–present; *exp. 2030*). Co-advised with Andreas Tolias.

Visiting Ph.D. Students

1. **Louis Van Langendonck**, UPC BarcelonaTech (3 months, 2026).
2. **Marco Montagna**, Sapienza Rome (3 months, 2026).

Postdoctoral Fellows

1. **Fatih Dinc** (2025–present). Ph.D.: Stanford.
   Awards: Richard C. DiPrima Prize, SIAM (2026; biennial, single recipient worldwide); KITP Postdoctoral Fellowship (2024).
2. **Guillermo B. Gil** (2024–present). Ph.D.: Universitat Politècnica de Catalunya (UPC).
   Awards: UPC Extraordinary Doctoral Award (2026); Arlequin AI Fellowship (2025); First Prize, International Topological Deep Learning Challenge (2023).
3. **Sarah Kushner** (2025–present). Ph.D.: University of Toronto.

Staff

1. **Luis Pereira**, Software Engineer (2024–present).
2. **Alexander West**, Undergraduate & Lab Manager (2024–present).

Alumni Placement. Out of my 3 postdoctoral alumni, 2 hold faculty positions, and 1 is a staff researcher. One also co-founded a venture-backed company.

1. **Sophia Sanborn** — Research Assistant Professor, Stanford Medical School; Co-Founder, Metamorphic ($70M funded).
2. **David Klindt** — Assistant Professor, Cold Spring Harbor Laboratory.
3. **Bongjin Koo** — Scientist, Simons Machine Learning Center, New York Structural Biology Center.

My visiting Ph.D. alumni include **Daniel Kunin** (visiting PhD student in my lab for two years, now Miller Postdoctoral Fellow at UC Berkeley).

Teaching

I teach the following undergraduate and graduate classes at **UCSB**, with growing enrollment:

- **ECE 139**: Probability and Statistics. *147 students (2026).*
- **ECE 3**: Python Programming for Science and Engineering. *92 → 138 students (2021–2025).*
- **ECE 278A**: Digital Image Processing. *21 → 29 students (2022–2026).*
- **ECE 594N**: Equivariant, Topological and Geometric Deep Learning. *14 → 26 students (2022–2026).*

I was previously a lecturer at **Stanford University**:

- Introduction to Statistical Methods: Precalculus *(2019).* Principal, sole instructor.
- Statistical Methods for Engineering and the Physical Sciences *(2018, 2019).* Principal, sole instructor.
- Non-Euclidean Methods in Machine Learning *(2020).* Guest lecturer.

Service

I co-founded the NeurReps workshop at NeurIPS (5 annual editions, 300–500 in-person attendees each year), organized 6 international workshops at top AI/ML venues, served as reviewer, grant panelist, member of scientific committees, and have served on multiple ECE department committees at UCSB.

Conference & Workshop Organization

1. Co-founder and lead senior organizer, NeurIPS NeurReps Workshop on Symmetry and Geometry in Neural Representations (2022, 2023, 2024, 2025, 2026) — **160+ accepted submissions and 300–500 in-person attendees each year**, and 20+ invited speakers across the five editions.
2. Co-organizer, NeurIPS 2026 Workshop on Bridging Optimal Transport, Learning and Structured Data: Toward Geometric Distributional Learning.
3. Organizer, WIDS Datathon in partnership with Kaggle, Global & University Editions (2025) — **6,000+ participants from 99 countries.**
4. Co-organizer, CVPR 2024 Workshop on Topological Deep Learning for Computer Vision.
5. Lead co-organizer, BIRS 2023 Workshop on Mathematical Methods for Morphological Shapes across Biological Scales.
6. Lead co-organizer, MICCAI 2022 Workshop on Topological Data Analysis and its Applications for Medical Data.
7. Lead organizer, ICLR 2022 Workshop on Geometrical and Topological Representation Learning.
8. Lead senior organizer of 6 AI/ML international challenges (2021-2026).

Peer Review & Scientific Committees

1. Reviewer for *Nature*, *Nature Communications*, *NeurIPS*, *ICML*, *ICLR*, *CVPR*, *KDD*, *JMLR*, *Foundations of Computational Mathematics*, *ACM Transactions on Mathematical Software*, *Transactions on Graphics*, *Journal of Mathematical Imaging and Vision*.
2. Grant panel reviewer, NSF (Unsolicited Panel, 2022) and Keck Foundation (2026).
3. Scientific Committee, Conference of Geometric Science of Information (GSI) — 2017, 2019, 2021, 2023, 2025.
4. Mentor, MIT Summer Geometry Institute (2021, 2022); London Geometry in Machine Learning Summer School (2022); Outreachy program for underrepresented populations (2021).

Department & University Service

1. Chair, ECE Public Relations Committee (2025–present).
2. Member, ECE Grants Strategy Committee (2025–present).
3. Member, ECE Seminars Committee (2022–2024).
4. Member, ECE Graduate Admissions (2021–2023).
5. Undergraduate Advisor, ECE (2021–2022).

Industry Advising & Consulting

1. Advisor, Atmo, Inc. (2021–present).
2. Advisor and Consultant, New Theory, Inc. (2024–present).
3. Consultant, Los Angeles Dodgers (2025).
4. Consultant, CaranX (2022–2024).

Outreach

I count eight years of public science communication (2017–present) reaching K–12 students, teachers, and general audiences.

1. Lead organizer, REAL AI for Science public lecture series — weekly in San Francisco (2017–2020), quarterly in Santa Barbara (2022–2025); 8 years of continuous public science communication.
2. Co-organizer of K–12 outreach events at UCSB for middle schoolers (De Anza Academy GRIT), high schoolers (St Therese Academy), and science teachers (MRL Workshop) — 90+ students and teachers reached (2023–2024).
3. Outreach speaker, Lower East Side Girls Club, New York City (2024).
4. Invited to give 7 outreach talks for UCSB (see Talks section).

5. Ambassador, L'Oréal-UNESCO For Women in Science (2022–present).

Press & Media

Bold: candidate ***Italic Bold***: advised Ph.D. student, visiting Ph.D. student, postdoc, or alumni

Op-Eds

1. *PLOS Biology.* "Artificial Scientific Intelligence: The Fifth Era of Science." **Miolane** (2025).
2. *Inside Higher Ed.* "Science Can't Depend on Corporate Goodwill." Lamar, **Miolane** (2025).

Press & Podcasts

1. *Nature.* "Brain differences between sexes get more pronounced from puberty." By Simms — coverage of Huang, Jamison, Jacobs, **Miolane**, Kuceyeski (2026).
2. *The Transmitter.* "Error equation predicts brain's ability to generalize." By Mesa — featuring expert commentary by **Miolane** on neural geometry and AI (2026).
3. *The Economist* (podcast). "We Don't Need No (AI) Education." ***Bertics***, Featuring **Miolane** (2025).
4. *The Transmitter.* "How Neuroscientists Are Using AI." Featuring **Miolane** (2025).
5. *Science et Vie.* "The Secrets of Our Thoughts Revealed by Mathematics." — coverage of **Miolane**'s research (2023).
6. *TWiML* (podcast). "Geometric Statistics in ML with Geomstats." — **Miolane**'s interview (2018).
7. *France Inter* (podcast, national radio). "Le Club de la Tête au Carré." Featuring **Miolane** (2016).